

DESi ablation Phase-3 — purpose-built lifecycle/status cases, two models

Real run · temp 0 · 3 reps · 2 cases where the chat CONTAINS superseded / ruled-out content (a reversed rollout + a ruled-out hypothesis; a deprecated field + a relaxed 'forever' rule). Retrieval surfaces the stale items; the curated DESi slice holds only the live state. Conservative; small N.

Recall (mean of reps) — Claude-Sonnet-4.5 / GPT-4o

case · model	A	B	C	D	E	F	G	H	R1	R2	R3
case_s1 · sonnet	0.867	1.0	0.0	1.0	1.0	0.0	0.0	0.2	0.4	0.4	0.4
case_s2 · sonnet	0.933	1.0	0.0	1.0	0.933	0.0	0.0	0.6	0.2	0.2	0.2
case_s1 · gpt-4o	0.6	1.0	0.0	1.0	1.0	0.0	0.0	0.067	0.4	0.4	0.4
case_s2 · gpt-4o	0.867	1.0	0.0	1.0	1.0	0.0	0.0	0.2	0.067	0.067	0.067

Headline: B (curated live state) vs retrieval & baselines — both models

comparison	sonnet mean Δ	gpt-4o mean Δ
B – R1	0.7	0.766
B – R2	0.7	0.766
B – A	0.1	0.267
B – F	1.0	1.0
B – C	1.0	1.0
B – E	0.033	0.0

Degeneration (mean across the 2 cases) — sonnet / gpt-4o

cond	loop s/g	invalid s/g	bad_frame s/g	conf_wrong s/g	self_conf s/g
A	0.0/0.0	0.0/0.0	0.0/0.0	0.0/0.0	93.5/95.8
B	0.0/0.0	0.0/0.0	0.0/0.0	0.0/0.0	93.5/100.0
C	0.0/0.0	15.0/14.334	1.0/1.0	1.0/1.0	93.5/100.0
D	0.0/0.0	0.0/0.0	0.0/0.0	0.0/0.0	88.5/100.0
E	0.0/0.0	0.0/0.0	0.0/0.0	0.0/0.0	90.15/100.0
F	0.5/0.666	0.0/0.0	0.0/0.0	1.0/0.5	95.0/83.35
G	0.0/0.0	8.0/8.0	1.0/1.0	1.0/1.0	95.0/100.0
H	1.0/0.334	0.833/0.666	0.834/0.666	0.666/1.0	75.0/100.0

cond	loop s/g	invalid s/g	bad_frame s/g	conf_wrong s/g	self_conf s/g
R1	0.0/0.0	0.0/0.0	0.0/0.0	1.0/1.0	79.5/95.0
R2	0.0/0.167	0.0/0.0	0.0/0.0	1.0/1.0	79.5/93.35
R3	0.0/0.167	0.0/0.0	0.0/0.0	1.0/1.0	79.5/93.35

Multi-turn invalid_claim_persistence (reuse t1→t2→t3 → verdict)

case	cond	sonnet	gpt-4o
case_s1	C	12→0→12 persisted	12→0→12 persisted
case_s1	H	1→0→0 corrected	0→0→0 not_adopted
case_s2	C	18→0→18 persisted	18→0→18 persisted
case_s2	H	2→2→0 corrected	0→0→0 not_adopted

Conservative findings

- 1. On lifecycle/supersession tasks, the curated DESi state beats retrieval — large and cross-model.** B=1.0 on both models; BM25/TF-IDF/hybrid retrieval = 0.07–0.4. B–R1 = 0.7 (sonnet) and 0.766 (gpt-4o), per-case 0.6–0.93. Retrieval surfaces the superseded items and the model reports them — confidently (retrieval conf_while_wrong = 1.0). This is the cleanest evidence that DESi's value here is a MAINTAINED epistemic state (supersession already resolved), not generic top-k selection.
- 2. CAVEAT (does not overclaim).** B's slice IS the curated ground-truth live state — DESi was handed the resolved state while retrieval had to resolve it. So this shows a correctly-maintained state beats retrieval on lifecycle tasks; it does NOT show DESi can BUILD that state automatically. That construction step is the real open question.
- 3. Poisoning + confidently-wrong replicate across models.** C invalid_reuse ≈14–15, G ≈8, bad_framing 1.0 on both; self-confidence stays ≈93–100 while wrong (gpt-4o reports 100 even when wrong). Multi-turn: the wrong slice (C) PERSISTS and even RELAPSES (12→0→12: dropped when probed, then reverted) on both models.
- 4. Metadata governance still not established.** B–E = 0.033 (sonnet) / 0.0 (gpt-4o): the live-state SELECTION carries the value, not the category/typing metadata.
- Supported:** correct state selection is load-bearing; plausible-wrong/irrelevant structure is actively toxic and sticky; a maintained DESi state beats retrieval on epistemic-status/lifecycle tasks (cross-model).
Not supported / open: automatic state construction; metadata governance; 'DESi makes the model smarter' / 'wins generally'.

Cost: ≈156 calls across both models. Power: N=2 cases is formally under-powered for a 0.1 effect, BUT the observed B–retrieval effect (~0.7) is ~5–7× the noise floor and replicates over 2 models × 2 cases (all points 0.6–0.93).
Next: more such cases (10–20) to tighten CIs; test automatic slice construction; a neural-embedding R2.